ENVIRONMENTAL PERFORMANCE, LCA AND ESG SUSTAINABILITY REPORT

Project: **Example - tutorial house in Cape Town (ERA5 1991-2020) [costs to be entered by the user]**

Life Cycle Assessment (LCA), Carbon Footprint and Sustainable Certifications

ISO 14040 / ISO 14044 · GHG Protocol

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Triton 1.0.0 · International Edition (INT)

CLIMATE LOCATION	CLIMATE BASIS	SERIES PERIOD	TARIFF LOCATION
Cape Town / South Africa	INM-CM4 · HIST	01/01/1991 a 31/12/2020	CUSTOM

TANK / SAVINGS	EMBODIED CARBON	AVOIDED CO2 / YR	20-YEAR NET CARBON	CARBON PAYBACK
1,000 L	**160.4 kg**	**+0.1 kg/yr**	**+162 kg**	**> 50 years**
9.5% potable replaced	Materials & Equipment	Net Annual Balance	Lifetime Net Mitigation	-0.39 g CO2e/L avoided

1. 1. BUILDING & DEMAND INPUTS

Parameter	Value
Rainfall Series	**CapeTown.csv**
Catchment Area (Roof)	**150.0 m²**
Runoff Coefficient (C)	0.85
First Flush Discard	2.0 mm
Occupants	4 occupants
Per-capita Demand	150 L/person/day
Rainwater Target Share	**28%**

2. 2. HYDRAULIC BALANCE (YAS MODEL)

Hydraulic Indicator	Result
Storage Tank Volume	**1,000 Litros**
Potable Savings	**9.5%**
Annual Pluvial Use	**20.8 m³/yr (57.1 L/day)**
Residual Potable Demand	198.3 m³/yr (542.9 L/day)
Annual Overflow	23.8 m³/yr (65.2 L/day)
Full Reliability Days	**29.7% of days**
Total Water Demand	219.2 m³/yr

3. 3. LIFE CYCLE INVENTORY (LCI — ISO 14040/14044)

Component / Phase	Quantity / Size	Embodied CO2	Op. CO2/Yr	20-Yr Balance	Source / Reference
Lower Cistern (Polyethylene)	1,000 L (22.0 kg)	**+35.5 kg**	--	**+35.5 kg**	Resin: Franklin Associates for the American Chemistry Council (out/2020), 'Cradle-to-Gate Life Cycle Analysis of High-Density Polyethylene (HDPE) Resin', Table 6: 1,612 kg CO2e per 1,000 kg of resin (TRACI 2.1 / IPCC 2013 GWP-100, US scope). Replaces the previous internal value of 3.00, which had no source and declared the same boundary. The mass per m^3 remains an internal scenario.
Rigid PVC Pipe (DN 32 mm)	20.0 m (0.71 kg CO2/m)	**+14.1 kg**	--	**+14.1 kg**	Manufacturer's data sheet (2025), solvent-weld pipe 32 x 6 m: 1.793,14 g × 2,36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3)
Rigid PVC Pipe (DN 40 mm)	5.0 m (1.01 kg CO2/m)	**+5.0 kg**	--	**+5.0 kg**	Manufacturer's data sheet (2025), solvent-weld pipe 40 x 6 m: 2.561,27 g × 2,36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3)
Rigid PVC Pipe (DN 100 mm)	20.0 m (2.78 kg CO2/m)	**+55.5 kg**	--	**+55.5 kg**	Standardised pipe geometry, reinforced series: OD 101,6 mm, wall 2,5 mm × 2,36 kg CO2e/kg (EPD S-P-05503, RXP PVC Pipes, A1-A3)
Pressurising Pump (0.24 CV - Cast Iron, Copper & Aluminium)	1 un. (Principal) — 0.24 CV	**+29.1 kg**	--	**+29.1 kg**	Manufacturer catalogue mass (60 Hz) × 4,77 kg CO2e/kg from the Grundfos EPD EPD-GRU-20250340-CBA1-EN (A1-A3): order-of-magnitude approximation; the EPD of the manufacturer actually used prevails
First-Flush Discard System (Leaf Filter + Tank 300 L)	300 L (9.3 kg material)	**+21.1 kg**	--	**+21.1 kg**	First-flush discard / Polymer & Stainless-steel inventory
Pumping Electricity Use	21.2 kWh/yr (1.02 kWh/m^3)	**--**	--	--	Editable grid factor; use the factor of the site and of the study period
Treatment Line	20.8 m^3/yr; 0,048 kg product/yr (1.50 mg Cl2/L; 0,004 kg CO2e/m^3)	**--**	**+0.1 kg/yr**	**+1.7 kg**	Stages: Chlorination (calcium hypochlorite tablet feeder). Sized by the Triton water-quality module
Mains Potable Water Avoided	20.8 m^3/yr (0.00 kg CO2/m^3)	**--**	--	--	Editable potable-water factor; replace with the inventory of the entity that supplies the site

4. 4. CARBON BALANCE, ECO-PAYBACK & METHODOLOGY

Total Embodied Carbon	Annual Pumping	Chlorination / Treatment	Avoided Water Credit	Net Annual Saved	Eco-Payback
+160.4 kg CO2e	**+0.0 kg/yr**	**+0.1 kg/yr (0,048 kg product/yr at 1.50 mg Cl2/L)**	**-0.0 kg/yr**	**+0.1 kg CO2e/yr**	**> 50 years**

Carbon Balance Formulation & Convention: *Official convention: Carbon emitted to atmosphere is POSITIVE (+) in red (environmental impact); carbon avoided/removed from atmosphere is NEGATIVE (-) in green (ecological benefit).*
20-Year Net Balance (kg CO2e) = Embodied CO2 (+) + [Pumping CO2 (+) + Treatment CO2 (+) - Avoided Grid Water (-)] × 20 yrs. Environmental payback represents the time required for operational avoided emissions to offset initial material embodied carbon.

Treatment train considered: Chlorination (calcium hypochlorite tablet feeder).

5. 5. ESG RATING & GREEN BUILDING CERTIFICATIONS

🏛️ LEED v4 / v4.1 (WE)

0 of 8 points (Max: 8 pts)

Scored 0/8 pts (Below the LEED threshold (min. 25%))

WEc1 (Ext.): 0/2 pts | WEc2 (Int.): 0/6 pts

🏛️ LEED v5 BD+C (WEc2)

2 of 8 points

Options 2 to 6

Option 2: 0 | 4: 0 | 6: 2 | 1: 0

🌍 EDGE (IFC)

Water bar: 9.5%

Short by 10.5 p.p.

No points: EDGE measures percentage against a local base case. WEM14 catchment area: 150 m^2

💧 GRI 303 & Water Positive

20.8 m^3/yr preserved

Ratio: 0.10x

417 m^3 over 20 yrs

🌿 ESG Executive Synthesis & Decarbonization

The project reaches an overall potable-water substitution of 9.5% (indoor: 9.5%, outdoor: 0.0%), saving 21 m^3/yr of treated water (417 m^3 accumulated over 20 years). Under LEED v4 WE (maximum of 8 points), the project scored 0 of 8 points. For EDGE, which awards no points but measures a percentage against a baseline case, the water bar reaches 9.5% — below the 20% threshold required in each of the three bars. The rainwater harvesting system plays a vital role in urban water security and climate resilience, contributing directly to Scope 3 GHG mitigation.

⚠️ Technical Responsibility & Methodological Notice:

This environmental report was generated by Triton following the applicable technical standards and the ISO 14040/14044 guidelines. Emission factors represent screening engineering scenarios and must be validated by the responsible environmental professional against supplier EPDs and the parameters of the local grid and water supply.